MACHINE LEARNING

Backpropagation Visual Guide

How neural networks learn:
forward pass, loss, chain rule, gradient descent.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Forward Pass

Input flows one way: the network makes a prediction.

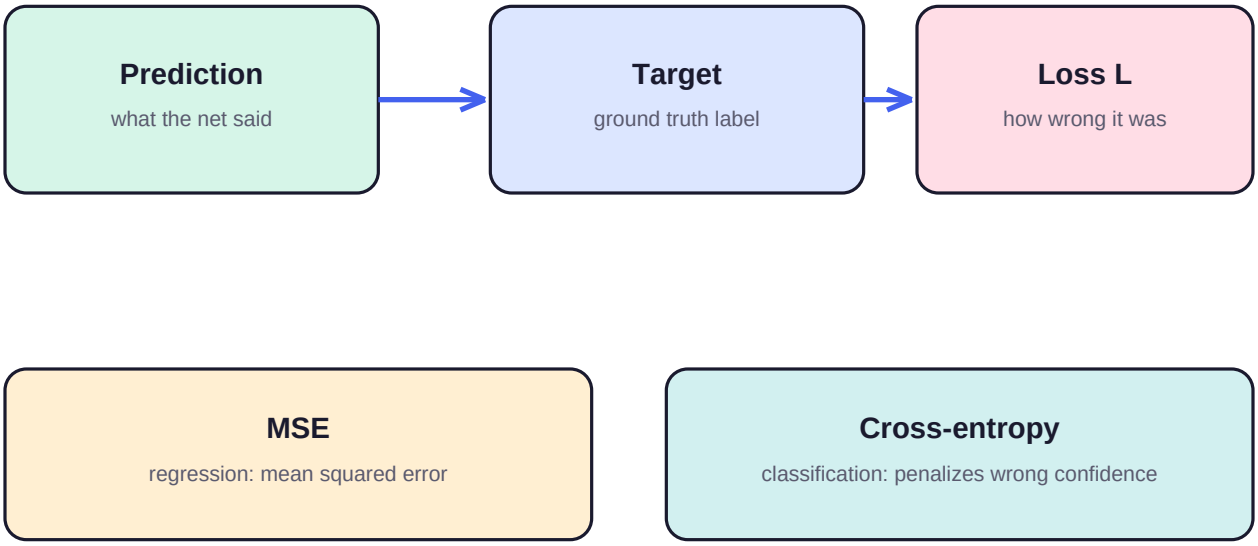
Each neuron: z = W dot x + b, then a = activation(z). Prediction flows left to right.

Forward pass = the network making a guess. Nothing is learned yet — learning comes from the error.

STEP 2/6

Measuring Error with Loss

One number says how wrong the prediction was.



Pick the loss that matches the task. The loss is the single number training tries to minimize.

STEP 3/6

The Chain Rule, Backwards

Blame is assigned layer by layer, in reverse.

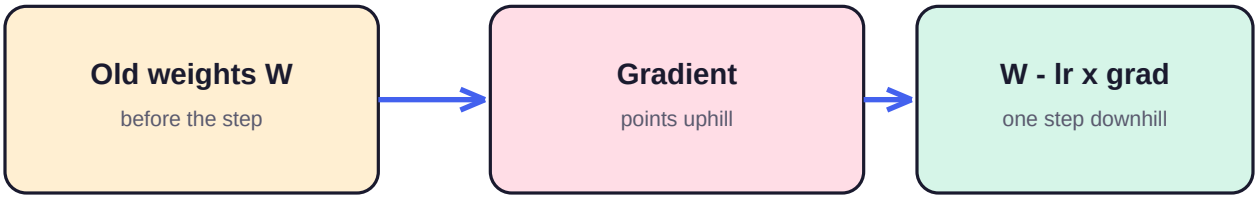
dL/dW = dL/dy x dy/dz x dz/dW — multiply local derivatives back through every layer.

The chain rule turns 'how wrong was I' into 'which weight caused it'. Gradients flow opposite to the forward pass.

STEP 4/6

Gradient Descent: The Update

Nudge every weight a little, opposite the gradient.



Update rule: W := W - lr * dL/dW. Subtract because the gradient points toward higher loss.

Repeat millions of times: forward, loss, backward, update. That loop IS training.

STEP 5/6

The Learning Rate

The single most influential training knob.

Too small

lr = 0.0001: crawls, may stall

Just right

lr = 0.01: steady descent

Too large

lr = 1.0: overshoots, diverges

Step size = learning rate x gradient. Too big bounces out of the valley; too small never arrives.

Start moderate (Adam default ~1e-3). Use a schedule: warm up, then decay as the loss plateaus.

STEP 6/6

Training in Practice

What the loop looks like at scale.

- Minibatches: update on 32-512 examples at a time — noisy gradients escape sharp minima.
- Epochs: one full pass over the data. Watch train AND validation loss — divergence means overfitting.
- Optimizers: SGD with momentum is simple; Adam/AdamW adapt the step per weight and are the default.
- Gradient clipping: cap gradient norms to stop explosions in deep or recurrent nets.
- Shuffle every epoch. Never tune hyperparameters on the test set.

If training is unstable: lower the learning rate first, check the data pipeline second, blame the architecture last.

STEP ✓/✓

Key Takeaways

Backpropagation — Remember

- Forward pass makes a prediction; loss measures the error as one number.
- The chain rule sends gradients backward, assigning blame to each weight.
- Gradient descent subtracts the gradient: W := W - lr * dL/dW.
- Learning rate controls step size — too large diverges, too small stalls.
- Minibatches + epochs + Adam is the standard training recipe; watch validation loss for overfitting.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev